WORKED SOLUTION

STEP II 1994, Question 3

Pure • Functional Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q3

The key idea

We know nothing about f except one identity, valid for *all* x and y, and one value, $f(0) = 1$. So the only tool is substitution: choose x and y so that most arguments become 0 or a, where we know (or are about to know) the values. The identity should remind you of $\cos(x-y) = \cos x \cos y + \sin x \sin y$, with $f(a-x)$ playing the part of $\pm \sin x$; that analogy tells you both what to expect and what the examples will be.

Solution

Throughout, write the given identity as

$$f(x-y) = f(x)\,f(y) - f(a-x)\,f(a+y) \qquad \text{for all } x, y. \tag{$*$}$$

Every step below is a substitution into $(*)$, so we make no assumption of continuity, differentiability or anything else.

Showing $f(a) = 0$

Put $x = y = 0$ in $(*)$:

$$f(0) = f(0)^2 - f(a)\,f(a), \qquad \text{so} \qquad 1 = 1 - f(a)^2.$$

Hence $f(a)^2 = 0$, and since $f(a)$ is a real number, $f(a) = 0$.

Part (i)

Put $x = 0$ and $y = t$ in $(*)$:

$$f(-t) = f(0)\,f(t) - f(a)\,f(a+t) = 1 \cdot f(t) - 0 = f(t),$$

using $f(0) = 1$ and $f(a) = 0$. So f is even.

Part (ii)

Put $x = y = a$ in $(*)$:

$$\mathrm{f}(0) = \mathrm{f}(a)^2 - \mathrm{f}(0)\,\mathrm{f}(2a), \qquad \text{so} \qquad 1 = 0 - \mathrm{f}(2a),$$

giving $\mathrm{f}(2a) = -1$.

Part (iii)

> **Ask yourself** Which single substitution makes the first product on the right vanish but keeps t free?

Setting $x = a$ kills the term $\mathrm{f}(x)\mathrm{f}(y)$ (because $\mathrm{f}(a) = 0$) and turns $\mathrm{f}(a - x)$ into $\mathrm{f}(0) = 1$, while y is still free. With $x = a$, $(*)$ becomes

$$\mathrm{f}(a - y) = 0 \cdot \mathrm{f}(y) - \mathrm{f}(0)\,\mathrm{f}(a + y) = -\mathrm{f}(a + y) \qquad \text{for all } y.$$

So f is "odd about a". Now choose $y = a - t$: then $a - y = t$ and $a + y = 2a - t$, so

$$\mathrm{f}(t) = -\mathrm{f}(2a - t),$$

which is $\mathrm{f}(2a - t) = -\mathrm{f}(t)$ for all t, as required. (Putting $t = 0$ recovers part (ii), a useful consistency check.)

Part (iv)

Apply (iii) twice, with (i) in between. First, (iii) with t replaced by $-t$ gives

$$\mathrm{f}(2a + t) = -\mathrm{f}(-t) = -\mathrm{f}(t), \tag{1}$$

using (i). So shifting the argument by $2a$ changes the sign. Shifting by $2a$ again, i.e. applying (1) with t replaced by $2a + t$,

$$\mathrm{f}(4a + t) = -\mathrm{f}(2a + t) = -(-\mathrm{f}(t)) = \mathrm{f}(t).$$

So f is periodic with period $4a$ (when $a \neq 0$), exactly like cosine with period $4 \cdot \frac{\pi}{2} = 2\pi$.

Examples

With $a = \pi/2$. Take $\mathrm{f}(x) = \cos x$. Then $\mathrm{f}(0) = 1$, and since $\cos(\frac{\pi}{2} - x) = \sin x$ and $\cos(\frac{\pi}{2} + y) = -\sin y$,

$$\mathrm{f}(x)\mathrm{f}(y) - \mathrm{f}(\tfrac{\pi}{2} - x)\mathrm{f}(\tfrac{\pi}{2} + y) = \cos x \cos y + \sin x \sin y = \cos(x - y) = \mathrm{f}(x - y).$$

With $a = -2$. Try $\mathrm{f}(x) = \cos kx$. The results above say we need $\mathrm{f}(a) = 0$, i.e. $\cos(2k) = 0$, and period $4|a| = 8$; so take $k = \frac{\pi}{4}$, i.e. $\mathrm{f}(x) = \cos\frac{\pi x}{4}$. To verify: $\mathrm{f}(0) = 1$, and with $u = \frac{\pi x}{4}$, $v = \frac{\pi y}{4}$,

$$\mathrm{f}(-2 - x) = \cos(-\tfrac{\pi}{2} - u) = \cos(\tfrac{\pi}{2} + u) = -\sin u, \qquad \mathrm{f}(-2 + y) = \cos(v - \tfrac{\pi}{2}) = \sin v,$$

so

$$\begin{aligned}\mathrm{f}(x)\mathrm{f}(y) - \mathrm{f}(-2-x)\mathrm{f}(-2+y) &= \cos u \cos v + \sin u \sin v \\ &= \cos(u-v) = \cos\tfrac{\pi(x-y)}{4} = \mathrm{f}(x-y).\end{aligned}$$

Both functions are clearly non-constant. (More generally, $\cos kx$ works for any a with $\cos ka = 0$, so $\cos\frac{3\pi x}{4}$ would also do for $a = -2$.)

Checks

Part (iii) at $t = 0$ gives $\mathrm{f}(2a) = -\mathrm{f}(0) = -1$, agreeing with (ii); and (iii) at $t = a$ gives $\mathrm{f}(a) = -\mathrm{f}(a)$, agreeing with $\mathrm{f}(a) = 0$. Numerically, for $\mathrm{f}(x) = \cos\frac{\pi x}{4}$ with $a = -2$, at $x = 1$, $y = 3$ the left-hand side is $\mathrm{f}(-2) = 0$ and the right-hand side is $\mathrm{f}(1)\mathrm{f}(3) - \mathrm{f}(-3)\mathrm{f}(1) = \cos\frac{\pi}{4}\cos\frac{3\pi}{4} - \cos\frac{3\pi}{4}\cos\frac{\pi}{4} = 0$. A computer testing both examples at 20 000 random pairs (x, y) found the identity holding to within 10^{-13}.

Common slips

- Getting the sign of the half-period shift wrong: (iii) and (i) give $\mathrm{f}(t+2a) = -\mathrm{f}(t)$, so the period is $4a$, not $2a$.
- Assuming f is differentiable (or is cosine) when proving (i)–(iv). The question forbids it; only substitutions are allowed.
- For $a = -2$, offering $\cos\frac{\pi x}{4}$ without checking that it satisfies the identity: matching $\mathrm{f}(a) = 0$ and the period is a good guide, not a proof.

Beyond STEP

Determining a function from an identity it satisfies is the subject of *functional equations*. Here one identity alone forces evenness, a "half-period" sign change and periodicity. In analysis courses one meets the classic results: Cauchy's equation $\mathrm{f}(x+y) = \mathrm{f}(x) + \mathrm{f}(y)$ forces $\mathrm{f}(x) = cx$ once we add continuity, and without such a condition there are wild solutions. In the same way the cosine subtraction formula characterises $\cos kx$ among continuous functions. For a careful treatment of how continuity enters such arguments, see Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.